

Numerical Astrophysics

Current affiliation:

Universidad de Sevilla
US, Sevilla, ESP

Former affiliations:

Pontificia Universidad Católica de Chile
PUC, Santiago, CHL

Universidad Autónoma de Madrid
UAM, Madrid, ESP

University of California
UCSC, Santa Cruz, USA

University of Graz
KF-Uni Graz, AUT

Position: Postdoctoral Researcher
Citizenship: Austrian, EU
CV updated: January 2026

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Experience & Skills

Programming Skills

Python

C & C++ • IDL • bash

SQL • JavaScript • LaTeX

machine learning • HPC

Language Skills

German (native)

English (C1)

Spanish (B2)

Soft Skills

Communication • Outreach

Improvisation Theatre • Dance

Leadership • Mentoring • Teaching

Academic Education

01/2015–12/2019*
UAM, Madrid, ESP
PhD in Theoretical Physics
Instituto de Física Teórica (IFT) & Universidad Autónoma de Madrid (UAM)

10/2005–03/2013
KF-Uni Graz, AUT
Diploma studies (Master) of Astrophysics
University of Graz

Awards & Fellowships

May 2025
Physicists of the Month
Austrian Physical Society

2023–2026
“The Formation Of Galaxies in the Large Scale Structure” (Co-PI)
PID20220-138855NB-C32
Ministerio de Ciencia e Innovación
200.000 €

2023–2025
“Modelización precisa de la radiación en el Universo” (Co-PI)
CNS2022-135878
Ministerio de Ciencia e Innovación
167.573 €

2023–2027
“LACEGAL-III” (travel funds)
HORIZON-MSCA-2021-SE-01
European Research Executive Agency and Innovation prog.
3.000 €

2022–2024
Margarita Salas Fellowship 2021 (PI)
CA1/RSUE/2021-00720
Ministerio de Universidades
121.100 €

2017–2022
“LACEGAL-II” (travel funds)
H2020-MSCA-RISE-2016
European Research Executive Agency and Innovation prog.
3.000 €

2015–2019
Severo Ochoa Doctoral Fellowship 2014 (PI)
SEV-2012-0249
Ministerio de Economía y Competitividad
93.150 €

Contracts & Professional Experiences

Jan.–June 2025
Sevilla, ESP
Postdoctoral Position in Numerical Astrophysics
University of Sevilla

2020–2021
Santa Cruz, CA, USA
Postdoctoral Position in Computational Astrophysics
University of California, Santa Cruz (UCSC)

April–Sept. 2013
Graz, AUT
Software Development
Freelance contractor at research institute
Joanneum Research

Further Education & Training (selected)

July–Aug. 2021
UCSC, Santa Cruz, USA
Equity-Minded Mentoring
Training on mentoring strategies and personal development
CITL

Oct. 25–27 2017
Madrid, ESP
Software Carpentry Workshop
Software Tools and Programming
IFT-CSIC/UAM

Sep. 29–Oct. 4 2016
La Plata, ARG
Special Course: SAM Galaxy Formation
Galaxy formation and evolution modelling incl. software tools
IALP

April 25–26 2015
Dublin, IRL
Women in High-Performance Computing
Initiation in high-performance computing and code parallelisation
PARC

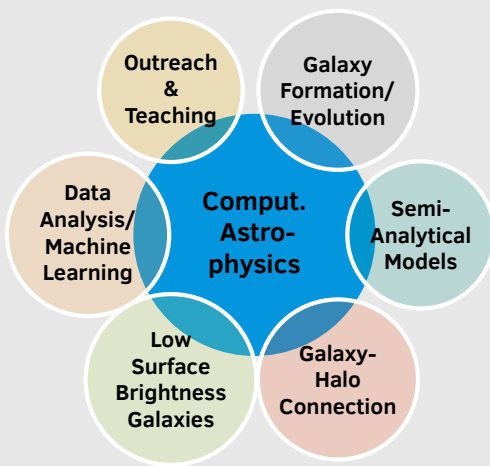
July 2011
Graz, AUT
Software Development
Internship at research institute
Joanneum Research

Teaching Experience

March–July 2023
Professor Astronomy (AST0211)
PUC, Santiago, CHL

Sept. – Nov. 2022
Ass. Prof. Astrofísica Experimental (AST0421)
PUC, Santiago, CHL

Research Interests



Project Overview (selected)

FOGLESS+The Machine 2025-pres.
Applying neural networks to cluster galaxy data (in collaboration with Universidad de Complutense, Madrid).
Involvement:
Status:

Hidden Figures in the Sky 2022-pres.
Unveiling the formation and assembly channels as well as the galaxy-halo connection of the in the literature underrepresented *low surface brightness galaxies* using various numerical models of galaxy formation and evolution.
Involvement:
Status:

A SAM Perspective on... 2017-pres.
Paper series focusing on different aspects of galaxy formation and evolution with semi-analytical models.
Involvement:
Status:

The Three Hundred 2017-pres.
Studying 324 central bright cluster galaxies and their evolutionary tracks and galaxy-halo connection through cosmic history.
Involvement:
Status:

- project leader
- main researcher
- support researcher
- data release
- machine learning
- pipelining/coding
- data analysis

Advised Theses



Nov. 2025-pres.	Master thesis Astrophysics A. Martínez Morata	VIU, Valencia, ESP
Sept.-Dec. 2022	Graduate Thesis Lic. Astrofísica S. Tancara	PUC, Santiago, CHL

Committees • Mentoring • Leadership



Oct. 2025-pres.	Mentor in SEA programme: <i>Mentoría para Astrónomos</i>	online
2024-pres.	Collaborator <i>Programa Autorreferentes: Mujer y Ciencia</i>	US
July 2023	Committee thesis Lic. Astrofísica V. Miranda	PUC, Santiago, CHL

Conferences & Workshop Activities (selected)



Dec. 9-14 2024 Puerto Varas, CHL	Deciphering the Cosmic Code for Galaxy Formation	Poster
Oct. 15-19 2024 Sevilla, ESP	X Meeting on Fundamental Cosmology	LOC
July 15-19 2024 Granada, ESP	XVI Reunión Científica de la Sociedad Española de Astronomía	Talk
Nov. 27-Dec. 1 2023 Montevideo, URU	XVII Latin American Regional IAU Meeting	Poster
2019, 2023 OAC, Córdoba, ARG	Friends of Friends Workshop Series	Talk
2019, 2022 & 2023	Galaxy-Halo Connection Collaboration Workshop Series PUC, Santiago, CHL, USM, Santiago, CHL, UDA, Copiapó, CHL	Talk
2019, 2020 & 2023	Mock Workshop Series OAC, Córdoba, ARG, UIBK, Innsbruck, AUT, ULS, La Serena, CHL	LOC, Invited Speaker, Talk

Seminars & Talks (selected)



28/04/2025 DIPC, Donostia, ESP	Hidden Figures in the Sky Evolution of low-surface brightness galaxies from a hydro-dynamical perspective	DIPC remote seminars
04/12/2024 ESO, Santiago, CHL	Three years of hiding Insights from studying low-surface-brightness galaxies in the EAGLE simulation within the <i>Hidden-Figures in the Sky</i> -project	TMT Seminar
	A SAM perspective on the challenges of modern galaxy formation modelling	
08/11/2022	Seminario del Departamento de Astronomía	ULS, La Serena, CHL
20/10/2022	Departamento de Astronomía Seminarios	Obs. Cerro Calán, CHL
27/09/2022	Instituto de Astrofísica Colloquia	PUC, Santiago, CHL
21/07/2021	Remote Astro Seminar Series	Uni. of Waterloo, CAN
05/10/2020 UCSC, Santa Cruz, USA	A SAM perspective on the assembly history and evolution of massive galaxies	UCSC Seminars

Further Scientific Activities



June-Dec. 2026 CAHA, Almería, ESP	Night observations with CAHA iKON 1.23m telescope Project TARSIS	4 nights
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Outreach Activities



 Sept. 2024 & 2025	La Noche Europea de l@s Investigad@res	📍 Sevilla, ESP
 2024-2025	Programa Autorreferentes: Mujer y Ciencia	📍 Sevilla, ESP
 July 2022	Novata por un Día: Visita guiada	📍 PUC, Santiago, CHL
 March 2020	Initiative: Astronomía desde el sofá → #AstroEnCasa	online
 2019-2021, 2025	Charla con una Astrónoma	online
 11/05/2019	IV Jornada de Astronomía en la UAM	📍 Madrid, ESP
 Nov. 5-15 2018	XVII Semana de la Ciencia y la Innovación (UAM)	📍 Madrid, ESP
 2017-2019	4 videos at the Youtube-channel @IFTMadrid with 124,000 views in total	📍 Madrid, ESP

Publications



In refereed journals:

- Stoppacher, D.**, P. Tissera, Y. Rosas-Guevara, G. Galaz, and J. Oñorbe (2025). “Hidden figures in the sky: Evolution of low-surface-brightness galaxies from a hydrodynamical perspective”. In: *A&A* 701, A272.
- Stoppacher, D.**, A. D. Montero-Dorta, M. C. Artale, A. Knebe, N. Padilla, A. J. Benson, and C. Behrens (2025). “A semi-analytical perspective on massive red galaxies: I. Assembly history, environment, and redshift evolution”. In: *A&A* 693, A219.
- Favole, G., V. Gonzalez-Perez, Y. Ascasibar, P. Corcho-Caballero, A. D. Montero-Dorta, A. J. Benson, J. Comparat, S. A. Cora, D. Croton, H. Guo, D. Izquierdo-Villalba, A. Knebe, Á. Orsi, **Stoppacher, D.**, and C. A. Vega-Martínez (2024). “Characterizing the ELG luminosity functions in the nearby Universe”. In: *A&A* 683, A46.
- Favole, G., V. Gonzalez-Perez, **Stoppacher, D.**, Á. Orsi, J. Comparat, S. A. Cora, C. A. Vega-Martínez, A. R. H. Stevens, C. Maraston, D. Croton, A. Knebe, A. J. Benson, A. D. Montero-Dorta, N. Padilla, F. Prada, and D. Thomas (2020). “[OII] emitters in MultiDark-Galaxies and DEEP2”. In: *MNRAS* 497.4, pp. 5432–5453.
- Stoppacher, D.**, F. Prada, A. D. Montero-Dorta, S. Rodríguez-Torres, A. Knebe, G. Favole, W. Cui, A. J. Benson, C. Behrens, and A. A. Klypin (2019). “A semi-analytical perspective on massive galaxies at $z \sim 0.55$ ”. In: *MNRAS* 486, pp. 1316–1331.
- Cui, Weiguang, Alexander Knebe, Gustavo Yepes, Frazer Pearce, Chris Power, Romeel Dave, Alexander Arth, Stefano Borgani, Klaus Dolag, Pascal Elahi, Robert Mostoghiu, Giuseppe Murante, Elena Rasia, **Stoppacher, D.**, and et al. (2018). “The Three Hundred project: a large catalogue of theoretically modelled galaxy clusters for cosmological and astrophysical applications”. In: *MNRAS* 480.3, pp. 2898–2915.
- Knebe, Alexander, **Stoppacher, D.**, Francisco Prada, Christoph Behrens, Andrew Benson, Sofia A. Cora, Darren J. Croton, Nelson D. Padilla, Andrés N. Ruiz, Manodeep Sinha, Adam R. H. Stevens, Cristian A. Vega-Martínez, Peter Behroozi, Violeta Gonzalez-Perez, Stefan Gottlöber, Anatoly A. Klypin, Gustavo Yepes, Harry Enke, Noam I. Libeskind, Kristin Riebe, and Matthias Steinmetz (2018). “MULTIDARK-GALAXIES: data release and first results”. In: *MNRAS* 474.4, pp. 5206–5231.

Further publications:

- Castrillo, A., S. Zamora, M. Rodríguez-Baras, M. Cortés-Contreras, I. Rebolledo, P. Corcho, I. Millán, **Stoppacher, D.**, R. M. Mérida, A. I. Díaz, and Y. Ascasibar (2020). “El observatorio astronómico de la UAM”. In: *Contributions to the XIV.0 Scientific Meeting (virtual) of the Spanish Astronomical Society*, p. 248.
- Cortés-Contreras, M., A. Castrillo, I. Rebolledo, M. Rodríguez-Baras, **Stoppacher, D.**, and S. Zamora-Arenal (2020). “Astro en casa”. In: *Contributions to the XIV.0 Scientific Meeting (virtual) of the Spanish Astronomical Society*, p. 253.